

Module 2

Task 2: Prompt Engineering Lab - Four-iteration refinement with Reflection

Prompt Engineering Iteration Log:

My target task (describe specifically — not 'emails' but the specific type, audience, and purpose):

Write a course overview summary for new healthcare ai course

VERSION 1 — Zero-shot prompt (write verbatim):

Write a course overview summary for a new course for engineering and healthcare students (asked topic, entered healthcare ai)

Output: Course Overview: AI in Healthcare

Output: This graduate course explores the design, implementation, and evaluation of artificial intelligence systems within clinical and healthcare settings, bridging technical foundations with real-world medical applications. Engineering students will deepen their understanding of machine learning, data pipelines, and system design as applied to diagnostic tools, predictive analytics, and clinical decision support. Healthcare students will gain the technical literacy needed to critically evaluate, implement, and collaborate on AI-driven solutions in patient care environments.

Topics include supervised and unsupervised learning for clinical data, natural language processing of medical records, algorithmic bias and fairness in healthcare AI, regulatory and ethical considerations (e.g., FDA approval pathways, HIPAA compliance), and case studies of deployed systems such as imaging diagnostics and risk-prediction models. Through interdisciplinary team projects, students from both engineering and healthcare backgrounds will collaborate to design a healthcare AI prototype, addressing both technical feasibility and clinical relevance.

By course end, students will be equipped to contribute meaningfully to AI development teams in healthcare, whether as technical architects or clinical stakeholders, with a shared vocabulary and mutual understanding of each other's domains.

V1 output quality (1–5): ____ . Two weaknesses I identified:

Quality - 3 – Added topics that were unexpected such as FDA approval pathways and 2). Called it a graduate course without prompting

VERSION 2 — +Role specification (write verbatim — include V1 plus new role addition):

You are an experienced team consisting of engineering and advanced practice nursing faculty at a research-intensive university who specializes healthcare ai, engineering and clinical practice, and teaching interdisciplinary graduate courses. Write a course overview summary for a new AI in healthcare course for engineering and healthcare students

Output: Course Overview: Artificial Intelligence in Healthcare — An Interdisciplinary Graduate Course

This graduate-level course is designed and co-taught by faculty in engineering and advanced practice nursing to prepare students from both disciplines to design, evaluate, and responsibly deploy AI systems in clinical environments. The course is built on the premise that effective healthcare AI requires not just technical sophistication but deep clinical grounding — and that the most consequential failures in this field tend to occur at the seams between the two.

Engineering students will build practical fluency in clinical workflows, healthcare data structures, and the regulatory and ethical landscape governing medical technologies, while applying their technical training to real clinical problems rather than abstracted datasets. Healthcare students — including advanced practice nursing and other clinical learners — will develop the technical literacy to critically appraise AI tools, participate meaningfully in their design, and lead their safe integration into practice. Neither group is treated as a passive consumer of the other's expertise; the course is structured so that clinical insight and engineering rigor inform each other from week one.

Core content includes machine learning fundamentals for clinical prediction and diagnosis, natural language processing applied to clinical documentation, algorithmic bias and health equity, human factors and clinician trust in AI-assisted decision-making, and the regulatory pathways (FDA, HIPAA, clinical validation standards) that govern real-world deployment. Throughout, case studies are drawn from deployed systems — including their documented failures — to ground discussion in the realities of clinical implementation rather than idealized performance metrics.

The course culminates in interdisciplinary team projects in which engineering and healthcare students co-design an AI-driven clinical tool, working through the full arc from clinical problem identification to technical prototyping to implementation planning. Teams are intentionally mixed by discipline, and grading rewards genuine collaboration over individual technical or clinical performance alone.

Graduates of this course will be equipped to function as effective bridge-builders in healthcare AI — whether as engineers who understand clinical context and constraints, or as clinicians who can engage critically and substantively with the technical teams building tools meant for their patients and practice settings.

V2 output quality (1–5): ____ . What improved vs. V1:

5 – Higher level language, more details, content and context.

VERSION 3 — +Few-shot example (paste or describe your example, then the prompt):

You are an experienced team consisting of engineering and advanced practice nursing faculty at a research-intensive university who specializes healthcare ai, engineering and clinical practice, and teaching interdisciplinary graduate courses. The course objectives include:

1. Be introduced to and practice explaining core concepts of artificial intelligence (AI).
2. Be introduced to, practice, and receive feedback on conducting AI assisted analysis of health data.
3. Be introduced to, practice, and receive feedback on evaluating ethical principles and standards in AI development and application.
4. Be introduced to, practice, and receive feedback on developing a Natural Language Processing (NLP) or Large Language Models (LLMs) supported tools for enhancing patient support or education.
5. Be introduced to and practice explaining how AI-driven tools support real-time feedback and clinical communication.
6. Be introduced to, practice, and receive feedback on integrating AI tools into a workflow to improve healthcare efficiency and support collaboration among healthcare professionals. Write a course overview summary for a new AI in healthcare course for engineering and healthcare students

Output: Course Overview: Artificial Intelligence in Healthcare — An Interdisciplinary Graduate Course

This graduate-level course, co-designed and co-taught by engineering and advanced practice nursing faculty, prepares engineering and healthcare students to work side by side on one of the most consequential challenges in modern healthcare: building AI tools that are technically sound, clinically meaningful, and ethically defensible. The course is structured around a "practice and feedback" model — students are introduced to core concepts, then apply them to real or realistic healthcare problems, then receive structured feedback from faculty with both technical and clinical expertise, mirroring the iterative, cross-checked process that responsible AI development demands in practice.

Students will first build a shared conceptual foundation in core AI principles, developing the ability to explain these concepts clearly to audiences with differing technical backgrounds — a skill essential to any interdisciplinary AI team. From there, the course moves into applied work: students will conduct AI-assisted analyses of health data, practicing the technical and interpretive judgment required to draw sound conclusions from clinical datasets. In parallel, students will critically evaluate the ethical principles and standards that govern AI development and use in healthcare, examining issues such as bias, fairness, transparency, and patient safety through real-world cases.

A central project thread runs through the course: students will design and develop a Natural Language Processing (NLP) or Large Language Model (LLM)-supported tool intended to

VERSION 3 — +Few-shot example (paste or describe your example, then the prompt):

enhance patient support or education, carrying it from concept through prototype with iterative faculty feedback. Alongside this, students will examine how AI-driven tools enable real-time feedback and support clinical communication, and will practice integrating AI tools into clinical workflows in ways that improve efficiency and strengthen collaboration among healthcare professionals.

Throughout, engineering and healthcare students are paired and grouped deliberately, ensuring that technical design decisions are continually informed by clinical context, and that clinical priorities are continually translated into technically actionable requirements. By the course's end, students will not only understand AI in healthcare conceptually — they will have practiced building it, evaluating it, and integrating it into real clinical and collaborative contexts, with direct feedback on each of these skills along the way.

V3 output quality (1–5): _____. What improved vs. V2:

More precise details pertaining to the course

VERSION 4 — +Chain-of-thought instruction (write full verbatim V4 prompt):

You are an experienced team consisting of engineering and advanced practice nursing faculty at a research-intensive university, with expertise in AI in healthcare, clinical practice, and teaching interdisciplinary graduate courses. The course objectives include:

1. Be introduced to, practice, and receive feedback on conducting AI assisted analysis of health data.

2. Be introduced to, practice, and receive feedback on evaluating ethical principles and standards in AI development.

3. Be introduced to, practice, and receive feedback on evaluating ethical principles and standards in AI development.
4. Be introduced to, practice, and receive feedback on developing a Natural Language Processing (NLP) tool for patient support or education.

5. Be introduced to and practice explaining how AI-driven tools support real-time feedback and clinical communication.

6. Be introduced to, practice, and receive feedback on integrating AI tools into a workflow to improve healthcare delivery.

professionals. Write a course overview summary for a new AI in healthcare course for engineering and healthcare students. What is the audience? (2) What is the single most important point? (3) What tone is appropriate? Then produce the overview.

Output: **Thinking it through:**

1. **Who is the audience?** Two distinct groups reading this: prospective graduate students (engineering and healthcare) who will enroll, and possibly curriculum committees or accreditation reviewers assessing course rigor and design. The course should be what they'll walk away able to do.
2. **What is the single most important point?** This isn't an engineering course with healthcare examples or a healthcare course with engineering examples. It's a genuinely interdisciplinary course where technical and clinical reasoning develop together, through hands-on practice.
3. **What tone is appropriate?** Authoritative but accessible — written by faculty who've done this work, for students who are not jargon-heavy; oriented around what students will *do*, not just what they'll *cover*.

Course Overview: Artificial Intelligence in Healthcare — An Interdisciplinary Graduate Course

This graduate course is co-taught by engineering and advanced practice nursing faculty and built for one purpose: to build, evaluate, and responsibly deploy AI tools in clinical practice. Rather than treating AI as either a purely technical or clinical topic, the course is structured around repeated cycles of practice and feedback, so students develop real competence — not just familiarity.

VERSION 4 — +Chain-of-thought instruction (write full verbatim V4 prompt):

Students begin by building a common foundation in core AI concepts, with practice explaining these ideas clearly. The team depends on. From there, the course moves into applied work: conducting AI-assisted analysis of real healthcare standards that shape responsible AI development and use, including bias, fairness, and patient safety.

A central project anchors the semester: in mixed engineering–healthcare teams, students design and develop a (LLM)-supported tool to enhance patient support or education, refining it through iterative faculty feedback from how AI-driven tools support real-time feedback and clinical communication, and practice integrating AI tools to strengthen collaboration among healthcare teams.

Every objective in this course is met through doing, not just covering — students leave having practiced the art of integration, with direct feedback on each, alongside engineering or healthcare peers who bring a different but e

V4 output quality (1–5): ____ . What improved vs. V3:

More professionally developed as if the response was ‘thought through.’

150-word reflective note — which technique produced the biggest jump? What remains unresolved?

Version 1 to version 2 produced a large jump, but still was not as detailed, targeted and polished as the output from prompts for version 3 to 4. Version 4 reflects professionalism, detail and a more ‘human voice.’ This was my first time using (Version 2) the role specification prompt. I usually wrote prompts specifying the target audience and not who was doing the writing. My previous Few shot prompts were mostly by trial and error over time and now I know that there is a term associated with this and providing something I have written provides the model better insight into what output I am seeking, This was my first time using Version 4 - Chain-of-thought and telling the model to think first. This is an exciting part of working with AI and I can pass this on to students and coworkers! Although this task took me a while to complete, I pushed myself to go back to other model platforms that I had tried previously. I am starting to like the output from Claude as well.